DevOps CI/CD with Jenkins
Software installation

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1 Compulsory Software

This software is required for the workshop. If you have problems installing, please contact me at:

victor.tan.33@gmail.com

1.1 Proper text editor

You will need a reasonably good text editor.

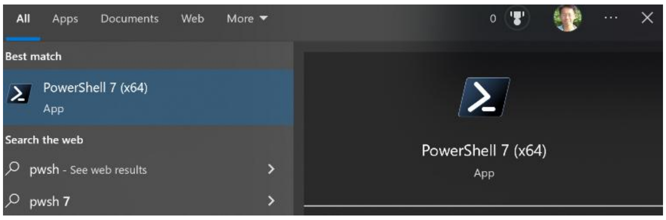
A good option for Windows is NotePad++.

If you have some alternative editor equivalent to NotePad++ which you are familiar and comfortable with, go ahead and use that instead.
Please **DON'T USE** the basic Notepad that comes with all Windows machines: this can cause problems with line formatting and alignment in certain parts of the workshop lab when we are editing material.

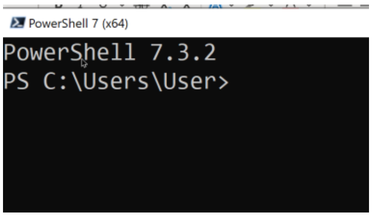
1.2 PowerShell 7

You can download and install it using the MSI package.

Once installed, search for `pwsh` in the Windows search box at the bottom left, and you should be able to see an icon for this app.



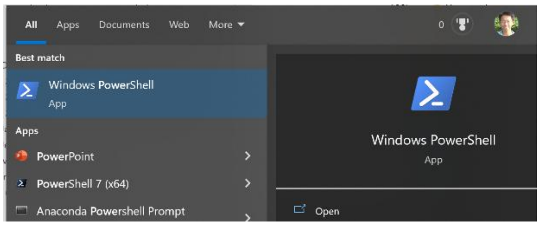
Once it starts up, you should see something like this:



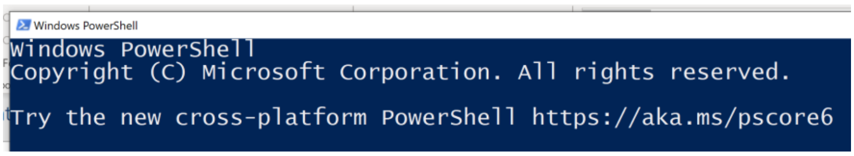
IMPORTANT NOTE

This PowerShell 7 is a different application from the standard PowerShell that comes with all Windows 10 installations - that one is known as PowerShell 5.1 or simply Windows PowerShell. PowerShell 7 is a cross-platform version (meaning it can run on BOTH Windows and Linux) of the original PowerShell 5.1.

Installing PowerShell 7 will NOT OVERWRITE PowerShell 5.1, you can still access PowerShell 5.1 by typing `powershell` in the Search box:



When it starts up, the shell will look like this:



1.3 MobaXTerm (SSH / SCP Windows client)

This is for establishing remote connection to Linux machines. If you already have another client that you frequently use (such as Putty) for this purpose and you are familiar with how to use that, then you don't have to install this.

Install and download from here. Make sure you download the **(Installer Edition) - GREEN BUTTON**

1.4 Suitable IDE (Visual Studio Code)

We need an IDE to inspect and modify the source code of the sample projects for this workshop. This includes source code written in .NET C#, Python and C++.

If you already have an IDE that can work with these languages, continue using that.

Otherwise, I recommend installing a widely used, popular open-source IDE: Visual Studio (VS) Code

1.5 Sign up for a Gmail account (if you don't have one)

NOTE: The reason we need a Gmail account for this workshop is because we are going to use Google's free SMTP server service. Therefore, you **CANNOT** use your company / organization email account or Outlook / Yahoo / other provider email account if you have one.
You **MUST** create a Gmail account, if you don't already have one. The two links below detail how to do this.

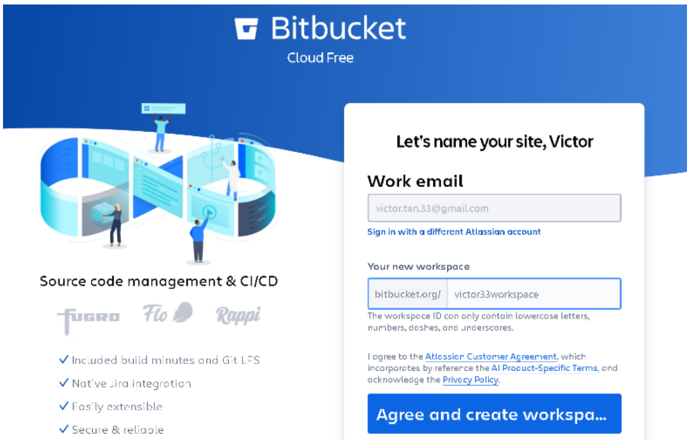
https://www.rightinbox.com/blog/how-to-create-a-new-gmail-account

https://support.google.com/mail/answer/56256?hl=en

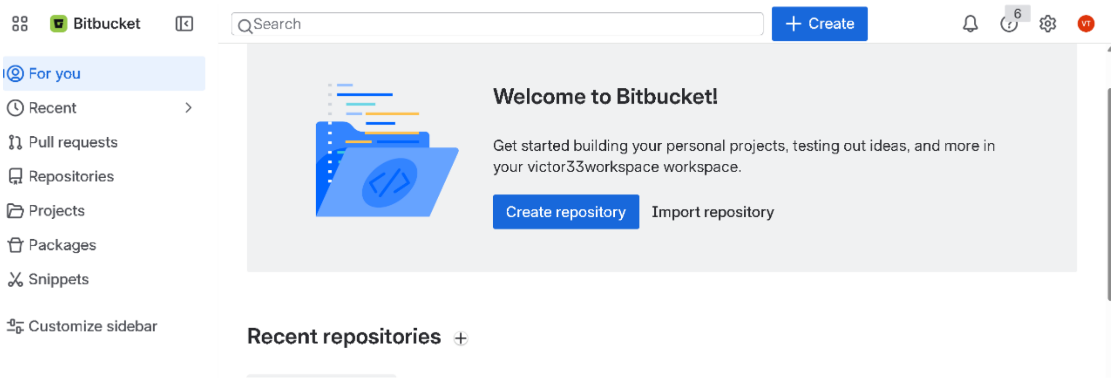
1.6 Sign up for a BitBucket account (if you don't have one)

At the main BitBucket page, you can click on Get It Free, after which you will be presented with an option to either login using any of your existing valid social media accounts, or optionally to create a completely new account on BitBucket.

Once you have logged in successfully, you may need to create a new workspace with a unique name to place all your repositories in.



Once you are done with that, you should be able to see the main page for that workspace.



Many videos on how to do this, this is an example:

https://www.youtube.com/watch?v=IoAqM1CSUHc

1.7 Sign up for a GitHub account (if you don't have one)

At the main GitHub page, you can click on Sign up, after which you will go through the process of creating a new Git Hub account.

Many videos on how to do this, this is an example:
https://www.youtube.com/watch?v=rflDMd5bkz4

1.8 Sign up for a DockerHub account (if you don't have one)

We will be using DockerHub to store the custom images created during our lab sessions.
You will need a valid DockerHub account for this purpose.

You can click Sign up at the official home page to obtain this.
Detailed instructions for creating your DockerHub account are also outlined here

2 Optional software installation on Windows

The workshop labs will primarily be conducted on Linux VMs (EC2 instances) running on AWS. This ensures that all required software for this workshop is properly installed and configured on these instances

However, some of you may wish to run some of the workshop lab exercises on your own machines, in addition to the cloud VMs. The software packages and their installation instructions are therefore provided below.

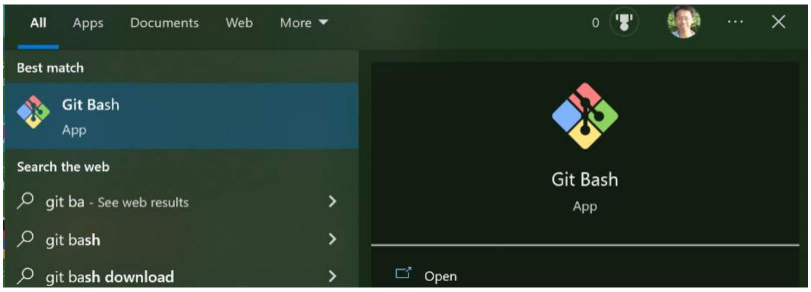
The installation instructions below assume you are using a **64-bit Windows machine** and that you have **full admin privilege** on the machine you are performing installation on. If you are using a company laptop where your account does not have full admin privilege, you may need to get your system admin to help perform the installation on the company laptop on your behalf.

2.1 Git

Download and [install from Git for Window] as it also already includes a BASH shell and GUI as well that we need for working with Git in Windows.

DO NOT use the official Git site because it does **NOT** include these additional tools which we need.

Once installed, search for `Git Bash` in the Windows search box at the bottom left, and you should be able to see an icon for this app.



When you start it, the Git Bash shell should look something like this:

If you cannot start the Git Bash shell as shown above, uninstall and repeat the installation process

2.2 Java 21 (or later)

NOTE: You must have **at least Java 21** installed as this is the **minimum requirement for the latest version of Jenkins.**

The official download site for Java lists all the various main versions of Java. If you don't already have Java 21 installed, then click on the link for any one of these versions (I would recommend Java SE 25).

You can download and install using the x64 installer for Windows.

A suitable step by step guide is available for Windows
Note: This guide assumes the latest version is Java 23 (`jdk-23`), so just replace all occurrences of the term `jdk-23` with `jdk-25` for Java 25.

Ensure that you add your Java installation `bin` directory to the `PATH` environment system variable. You should use the `JAVA_HOME` environment variable to add this directory to the `PATH` system variable, as outlined in these 2 guides below (change your `jdk` version appropriately when following them):
https://mkyong.com/java/how-to-set-java_home-on-windows-10/
https://www.baeldung.com/java-home-on-windows-mac-os-x-linux

A complete video guide for this process: https://youtu.be/Ex-T0cRjeIM
Note: This guide is for Java 21 (`jdk-21`), so just replace all occurrences of the term `jdk-21` with `jdk-25` for Java 25.

2.3 Jenkins

The official reference for installing Jenkins on Windows.

An online tutorial that is similar to this.

Some key points to note for the installation process using the Windows MSI installer:

a) For Step 3: Service Logon Credentials prompt, to simplify the setup choose the Logon Type as **Run the service as LocalSystem**. Although this is not recommended in the official guidelines from a security viewpoint, this is easier to setup than running Jenkins service as a local or domain user.

b) If you wish to follow the recommended official guideline of using the more secure and recommended option of running the service as a local or domain user, you will probably need to use a local user account (preferably with admin privilege) and enable service logon for that user account.

- Creating a local user account and changing it to admin account
- Enabling service logon

  You will also need to be able to access the home directory of the user account that you specified when going through the post installation set up wizard in order to unlock Jenkins.

c) **Install all suggested plugins** during the installation phase

d) **Make sure you note down** the admin username and password for accessing Jenkins: keep this simple so you can remember it, for e.g having the username and password being the same: `admin`

2.4 .NET SDK 10

Download the latest version of the SDK for .NET 10 to build apps
For a standard 64-bit Windows machine, select the Windows 64 installer.

2.5 Python

If you already have Python 3.12 or later installed on your machine that is adequate.

Otherwise, download a recent version of Python for Windows, using the standalone installer.